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Assignment #1: Software IP, Copyright and Ethics

Part 1: Concepts and Definitions

1. What is Intellectual Property and Copyright?

Intellectual Property (IP) refers to creations of the mind. It is a category of property that includes intangible creations of the human intellect. Common types of IP include inventions, literary and artistic works, designs, symbols, names, and images used in commerce. The law grants the creator exclusive rights to these creations for a certain period, allowing them to benefit from their work.

Copyright is a specific form of Intellectual Property protection. It is a legal right that grants the creator of an original work (such as a book, song, movie, or software code) exclusive rights to its use and distribution. A key characteristic of copyright is that it protects the **expression** of an idea, not the idea or concept itself.

2. How does software copyright protect Intellectual Property?

Software copyright primarily protects the literal **source code and object code** written by the developers.

- **Protection against copying:** It gives the owner the legal right to prevent others from copying, pasting, distributing, or selling the code without permission.
- **Derivative works:** It prevents others from modifying the code to create a new program based on the original work.
- **Limitations:** Crucially, copyright **does not** protect the underlying functionality, the algorithm, or the "idea" of what the software does. If a competitor writes completely new code that performs the exact same function, they have not violated copyright.

3. Differentiate Software Copyright from Patent

While both protect Intellectual Property, they cover different aspects of software and offer different levels of protection.

Feature	Software Copyright	Software Patent
What it Protects	The specific written code (the "expression" or text).	The underlying invention, process, method, or algorithm (the "functionality").
Acquisition	Usually automatic upon creation (though registration is helpful for lawsuits).	Requires a formal, expensive, and lengthy application process with a government patent office.
Duration	Very long (often the author's life plus 70 years).	Much shorter (usually 20 years from the filing date).
Scope	Narrow. It only stops people from stealing your specific code.	Broad. It stops people from using your method/algorithm, even if they write their own unique code for it.

Part 2: Discussion & Scenario

Scenario:

A software developer, Sarah, works for a small startup company that creates mobile applications. Sarah develops a unique algorithm for image processing that significantly improves the speed and quality of their flagship app. The company does not have a formal process for protecting intellectual property. Sarah leaves the company to start her own business and uses the same algorithm in her new app.

Questions and Answers:

1. What intellectual property issues arise in this scenario?

- **Ownership of "Work Made for Hire":** The primary issue is the legal ownership of the algorithm. In most jurisdictions, if an employee creates a work (such as code or an algorithm) within the scope of their employment and during work hours, the **employer** is considered the legal author and owner. This applies even if there is no specific contract

stating so. Therefore, Sarah likely does not own the IP she created while being paid by the startup.

- **Trade Secret Misappropriation:** Since the company did not patent the algorithm, it may be considered a "Trade Secret" (confidential information that gives the business a competitive edge). If Sarah utilizes confidential knowledge, she gained during her employment to aid a direct competitor (her own new business), this constitutes misappropriation of trade secrets.
- **Copyright Infringement:** If Sarah copied the actual source code she wrote at the startup and pasted it into her new app, she is infringing on the startup's copyright. However, if she rewrote the code from scratch but used the same logic, the issue shifts from copyright infringement to trade secret theft.

2. What steps could the startup company have taken to protect its intellectual property?

- **Employment Contracts with IP Assignment:** The company should have required Sarah to sign a contract explicitly stating that all inventions, code, designs, and algorithms created during her employment are the sole property of the company.
- **Non-Disclosure Agreements (NDAs):** An NDA would have legally bound Sarah to keep proprietary algorithms and business secrets confidential, even after her employment ended.
- **Non-Compete Agreements:** Depending on local labor laws, a non-compete clause could have prevented Sarah from starting a directly competing business or working for a competitor for a specific period after leaving the startup.
- **Patent Application:** If the algorithm was truly unique and novel, the company could have filed for a patent. A patent would have protected the functionality of the algorithm, preventing Sarah (or anyone else) from using that specific method, regardless of how the code was written.

3. What are Sarah's potential liabilities?

- **Civil Lawsuits and Damages:** The startup could sue Sarah for damages. If a court finds she stole Trade Secrets or infringed on Copyright, she could be forced to pay the startup for their lost profits or for the unjust enrichment she gained from using their IP.
- **Injunctions:** The court could issue an injunction (a legal order) forcing Sarah to stop using the algorithm immediately. This could effectively shut down her new business or force her to rebuild her application from scratch without the algorithm.
- **Breach of Fiduciary Duty:** Even without a written contract, employees generally have a "duty of loyalty" to their employer. Using company resources or time to build a rival product or asset can be seen as a breach of this duty, leading to further legal penalties.